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Modul 5

- MELAKUKAN GIT COMMIT PADA PROJECT YANG DIBUAT

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.22631.5832]
(c) Microsoft Corporation. All rights reserved.

C:\Users\asus>cd C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>cd C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git init
Initialized empty Git repository in C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics\.git\

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git add .

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git commit -m "Commit pertama - inisialisasi project"
[master (root-commit) 82a6d7c] Commit pertama - inisialisasi project
1 file changed, 1 insertion(+)
 create mode 100644 .nd

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git remote add origin https://github.com/rengganistan/HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02.git

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git branch -M main

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git push -u origin main
To https://github.com/rengganistan/HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/rengganistan/HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>cd "C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics"

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git pull origin main --rebase
remote: Enumerating objects: 57, done.
remote: Counting objects: 100% (57/57), done.
remote: Compressing objects: 100% (51/51), done.
remote: Total 57 (delta 21), reused 11 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (57/57), 19.34 MiB | 58.00 MiB/s, done.
From https://github.com/rengganistan/HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
dropping 82a6d7c8c9377c71f51b8e78d18a64209af41655 Commit pertama - inisialisasi project -- patch contents already upstream
Successfully rebased and updated refs/heads/main.

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git add .

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git commit -m "Resolve merge conflict"
On branch main
nothing to commit, working tree clean

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>
```

- MEMBUAT BRANCH BARU PADA GIT PROJECT

```
C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>cd "C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics"

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git branch implementasi-generic-method

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git checkout implementasi-generic-method
Switched to branch 'implementasi-generic-method'

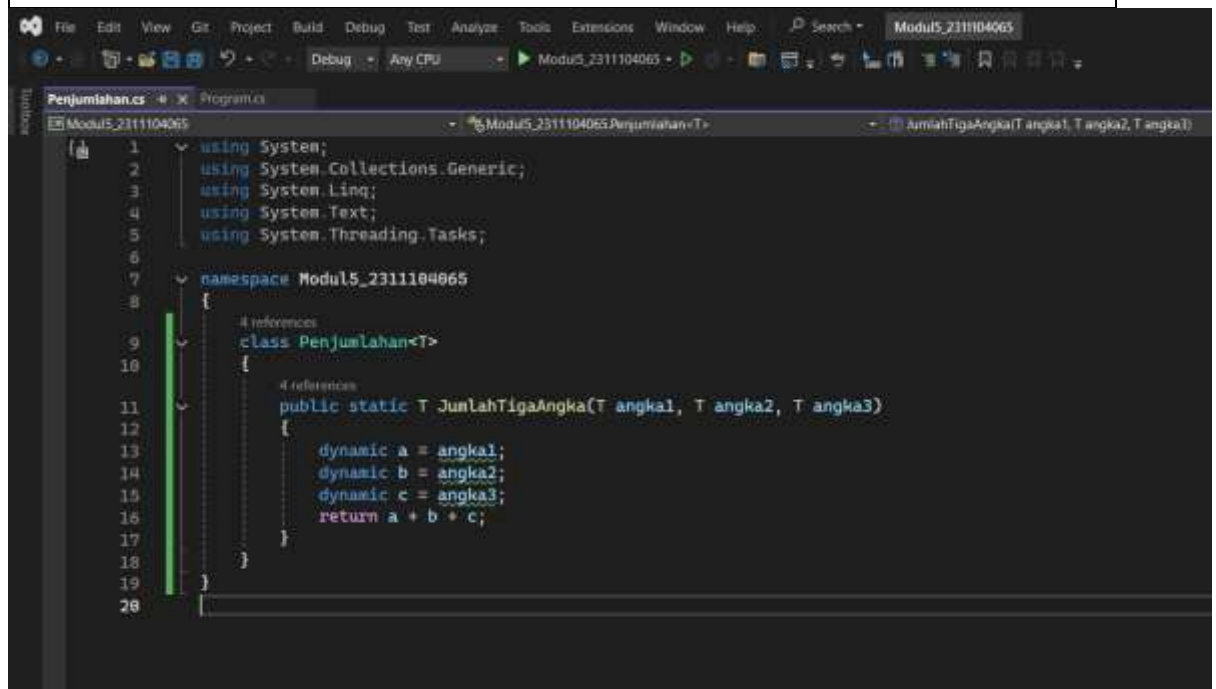
C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>git branch
* implementasi-generic-method
  main

C:\kpll_Rengganis Tantri Pramudita_2311104065_515E-07-02\05_Generics>
```

- MENAMBAHKAN METHOD DENGAN GENERIC

1. Penjumlahan.cs

```
namespace Modul5_2311104065
{
    class Penjumlahan<T>
    {
        public static T JumlahTigaAngka(T angka1, T angka2, T angka3)
        {
            dynamic a = angka1;
            dynamic b = angka2;
            dynamic c = angka3;
            return a + b + c;
        }
    }
}
```



2. Program.cs

```
using Modul5_2311104065;  
  
class Program
```

```
{
    static void Main()
    {
        // Masukkan NIM di sini
        string NIM = "12345678";

        // Ambil tiga angka dari NIM
        string angka1 = NIM.Substring(0, 2); // "12"
        string angka2 = NIM.Substring(2, 2); // "34"
        string angka3 = NIM.Substring(4, 2); // "56"

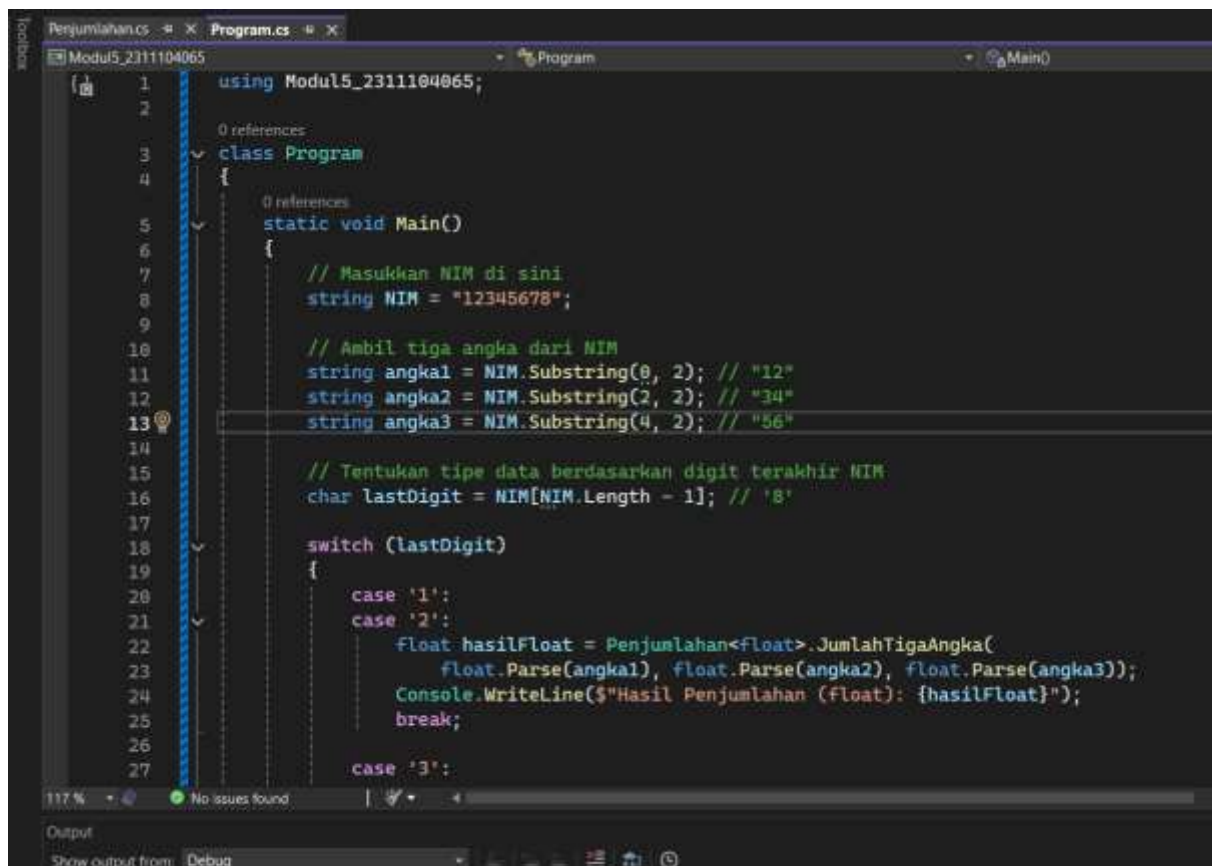
        // Tentukan tipe data berdasarkan digit terakhir NIM
        char lastDigit = NIM[NIM.Length - 1]; // '8'

        switch (lastDigit)
        {
            case '1':
            case '2':
                float hasilFloat = Penjumlahan<float>.JumlahTigaAngka(
                    float.Parse(angka1), float.Parse(angka2), float.Parse(angka3));
                Console.WriteLine($"Hasil Penjumlahan (float): {hasilFloat}");
                break;

            case '3':
            case '4':
            case '5':
                double hasilDouble = Penjumlahan<double>.JumlahTigaAngka(
                    double.Parse(angka1), double.Parse(angka2), double.Parse(angka3));
                Console.WriteLine($"Hasil Penjumlahan (double): {hasilDouble}");
                break;

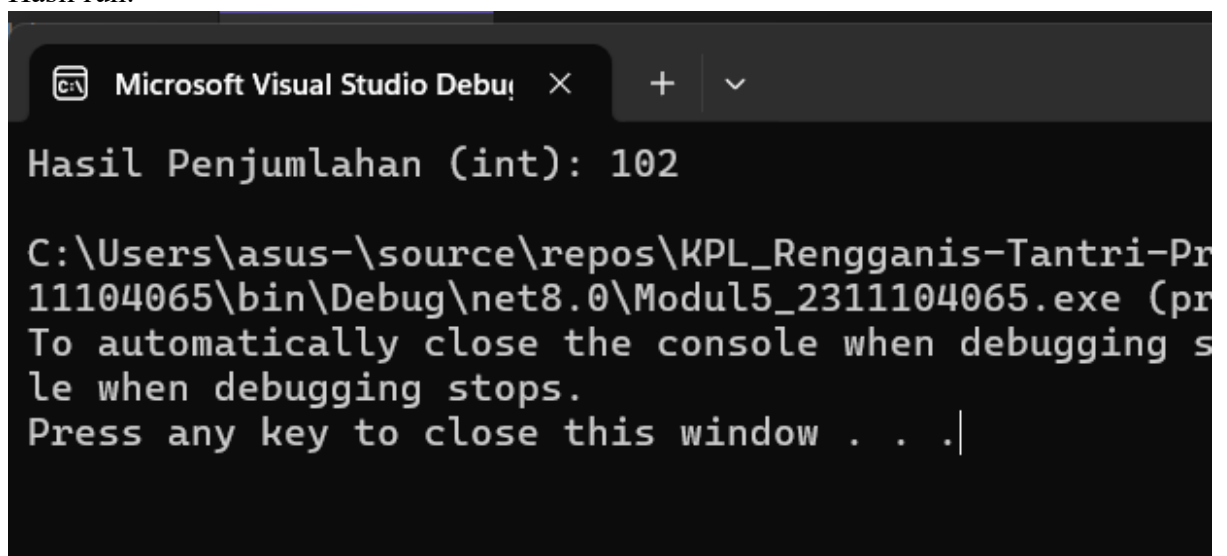
            case '6':
            case '7':
```

```
case '8':  
    int hasilInt = Penjumlahan<int>.JumlahTigaAngka(  
        int.Parse(angka1), int.Parse(angka2), int.Parse(angka3));  
    Console.WriteLine($"Hasil Penjumlahan (int): {hasilInt}");  
    break;  
  
case '9':  
case '0':  
    long hasilLong = Penjumlahan<long>.JumlahTigaAngka(  
        long.Parse(angka1), long.Parse(angka2), long.Parse(angka3));  
    Console.WriteLine($"Hasil Penjumlahan (long): {hasilLong}");  
    break;  
  
default:  
    Console.WriteLine("NIM tidak valid.");  
    break;  
}  
}  
}
```



```
1 using Modul5_2311104065;
2
3 class Program
4 {
5     static void Main()
6     {
7         // Masukkan NIM di sini
8         string NIM = "12345678";
9
10        // Ambil tiga angka dari NIM
11        string angka1 = NIM.Substring(0, 2); // "12"
12        string angka2 = NIM.Substring(2, 2); // "34"
13        string angka3 = NIM.Substring(4, 2); // "56"
14
15        // Tentukan tipe data berdasarkan digit terakhir NIM
16        char lastDigit = NIM[NIM.Length - 1]; // '8'
17
18        switch (lastDigit)
19        {
20            case '1':
21            case '2':
22                float hasilFloat = Penjumlahan<float>.JumlahTigaAngka(
23                    float.Parse(angka1), float.Parse(angka2), float.Parse(angka3));
24                Console.WriteLine($"Hasil Penjumlahan (float): {hasilFloat}");
25                break;
26
27            case '3':
```

Hasil run:



```
Microsoft Visual Studio Debug Console
Hasil Penjumlahan (int): 102

C:\Users\asus-\source\repos\KPL_Rengganis-Tantri-Pr
11104065\bin\Debug\net8.0\Modul5_2311104065.exe (pr
To automatically close the console when debugging s
le when debugging stops.
Press any key to close this window . . .|
```

- MELAKUKAN COMMIT, PUSH DAN PINDAH BRANCH

```
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>git add .
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>git commit -m "Menambahkan class Penjumlahan dengan method JumlahTigaAngka"
[implementasi-generic-method 944cda8] Menambahkan class Penjumlahan dengan method JumlahTigaAngka
 2 files changed, 74 insertions(+), 2 deletions(-)
 create mode 100644 Modul5_2311184865\Modul5_2311184865\Penjumlahan.cs
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>git push origin implementasi-generic-method
Enumerating objects: 9, done.
Counting objects: 100% (0/0), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.21 MiB | 621.89 MiB/s, done.
Total 0 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'implementasi-generic-method' on GitHub by visiting:
remote:   https://github.com/rengganistan/WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02/pull/new/implementasi-generic-method
remote:
To https://github.com/rengganistan/WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02.git
 * [new branch]      implementasi-generic-method -> implementasi-generic-method
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>git status
On branch implementasi-generic-method
nothing to commit, working tree clean
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>git checkout main
Switched to branch 'main'
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>git branch implementasi-generic-class
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>git branch
  implementasi-generic-class
  implementasi-generic-method
* main
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>git checkout implementasi-generic-class
#
Modul5_2311184865\Modul5_2311184865\Program.cs
Switched to branch 'implementasi-generic-class'
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>git branch
* implementasi-generic-class
  implementasi-generic-method
  main
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>git push -u origin implementasi-generic-class
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'implementasi-generic-class' on GitHub by visiting:
remote:   https://github.com/rengganistan/WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02/pull/new/implementasi-generic-class
remote:
To https://github.com/rengganistan/WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02.git
 * [new branch]      implementasi-generic-class -> implementasi-generic-class
branch 'implementasi-generic-class' set up to track 'origin/implementasi-generic-class'.
C:\Users\asus\source\repos\WPL_Rengganis-Tantri-Pramudita_2311184865_515E-07-02\05_Generics>
```

- MENAMBAHKAN METHOD DENGAN GENERIC

1. SimpleDataBase.cs

```
namespace Modul5_2311184865
{
    class SimpleDataBase<T>
    {
        private List<T> storedData;
        private List<DateTime> inputDates;

        public SimpleDataBase()
        {
            storedData = new List<T>();
            inputDates = new List<DateTime>();
        }

        public void AddNewData(T item)
        {
            storedData.Add(item);
            inputDates.Add(DateTime.UtcNow);
        }

        public void PrintAllData()
        {

```

```

        for (int i = 0; i < storedData.Count; i++)
        {
            Console.WriteLine($"Data {i + 1} berisi: {storedData[i]}, yang
disimpan pada waktu UTC: {inputDates[i]}");
        }
    }
}

```

```

7  namespace Modul5_2311104065
8  {
9      0 references
10     class SimpleDataBase<T>
11     {
12         private List<T> storedData;
13         private List<DateTime> inputDates;
14
15         4 references
16         public SimpleDataBase()
17         {
18             storedData = new List<T>();
19             inputDates = new List<DateTime>();
20         }
21
22         12 references
23         public void AddNewData(T item)
24         {
25             storedData.Add(item);
26             inputDates.Add(DateTime.UtcNow);
27         }
28
29         4 references
30         public void PrintAllData()
31         {
32             for (int i = 0; i < storedData.Count; i++)
33             {
34                 Console.WriteLine($"Data {i + 1} berisi: {storedData[i]}, yang disimpan pada waktu UTC: {inputDates[i]}");
35             }
36         }
37     }
38 }

```

2. Program.cs

```

class Program
{
    static void Main()
    {
        string NIM = "2311104065";

        string angka1 = NIM.Substring(0, 2); // "12"
        string angka2 = NIM.Substring(2, 2); // "34"
        string angka3 = NIM.Substring(4, 2); // "56"

        char lastDigit = NIM[NIM.Length - 1];

        switch (lastDigit)
        {
            case '1':
            case '2':
                SimpleDataBase<float> databaseFloat = new SimpleDataBase<float>();
                databaseFloat.AddNewData(float.Parse(angka1));
                databaseFloat.AddNewData(float.Parse(angka2));
                databaseFloat.AddNewData(float.Parse(angka3));
                databaseFloat.PrintAllData();
                break;

            case '3':
            case '4':

```

```

        case '5':
            SimpleDataBase<double> databaseDouble = new
SimpleDataBase<double>();
            databaseDouble.AddNewData(double.Parse(angka1));
            databaseDouble.AddNewData(double.Parse(angka2));
            databaseDouble.AddNewData(double.Parse(angka3));
            databaseDouble.PrintAllData();
            break;

        case '6':
        case '7':
        case '8':
            SimpleDataBase<int> databaseInt = new SimpleDataBase<int>();
            databaseInt.AddNewData(int.Parse(angka1));
            databaseInt.AddNewData(int.Parse(angka2));
            databaseInt.AddNewData(int.Parse(angka3));
            databaseInt.PrintAllData();
            break;

        case '9':
        case '0':
            SimpleDataBase<long> databaseLong = new SimpleDataBase<long>();
            databaseLong.AddNewData(long.Parse(angka1));
            databaseLong.AddNewData(long.Parse(angka2));
            databaseLong.AddNewData(long.Parse(angka3));
            databaseLong.PrintAllData();
            break;

        default:
            Console.WriteLine("NIM tidak valid.");
            break;
    }
}
}

```

```

3  class Program
4  {
5      0 references
6      static void Main()
7      {
8          string NIM = "2311104065";
9
10         string angka1 = NIM.Substring(0, 2); // "12"
11         string angka2 = NIM.Substring(2, 2); // "34"
12         string angka3 = NIM.Substring(4, 2); // "56"
13
14         char lastDigit = NIM[NIM.Length - 1];
15
16         switch (lastDigit)
17         {
18             case '1':
19             case '2':
20                 SimpleDataBase<float> databaseFloat = new SimpleDataBase<float>();
21                 databaseFloat.AddNewData(float.Parse(angka1));
22                 databaseFloat.AddNewData(float.Parse(angka2));
23                 databaseFloat.AddNewData(float.Parse(angka3));
24                 databaseFloat.PrintAllData();
25                 break;
26
27             case '3':
28             case '4':
29             case '5':
30                 SimpleDataBase<double> databaseDouble = new SimpleDataBase<double>();
31                 databaseDouble.AddNewData(double.Parse(angka1));

```

Hasil run:

```

Microsoft Visual Studio Debu  X  +  v
Data 1 berisi: 23, yang disimpan pada waktu UTC: 19/03/2025 06:14:04
Data 2 berisi: 11, yang disimpan pada waktu UTC: 19/03/2025 06:14:04
Data 3 berisi: 10, yang disimpan pada waktu UTC: 19/03/2025 06:14:04

C:\Users\asus-\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_11104065\bin\Debug\net8.0\Modul5_2311104065.exe (process 6028) exited
To automatically close the console when debugging stops, enable Tools->
le when debugging stops.
Press any key to close this window . . .

```

- MELAKUKAN COMMIT, PUSH DAN PINDAH BRANCH BAGIAN 2

```

C:\Users\asus\source\repos\HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02\05_Generics>git branch
* implementasi-generic-class
  implementasi-generic-method
  main

C:\Users\asus\source\repos\HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02\05_Generics>git add .

C:\Users\asus\source\repos\HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02\05_Generics>git commit -m "menambahkan class SimpleDataBase"
error: pathspec 'class' did not match any file(s) known to git
error: pathspec 'SimpleDataBase' did not match any file(s) known to git

C:\Users\asus\source\repos\HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02\05_Generics>git commit -m "menambahkan class SimpleDataBase"
[implementasi-generic-class 65824ec] menambahkan class SimpleDataBase
3 files changed, 112 insertions(+), 2 deletions(-)
create mode 100644 Modul5_2311104065/Modul5_2311104065/Penjumlahan.cs
create mode 100644 Modul5_2311104065/Modul5_2311104065/SimpleDataBase.cs

C:\Users\asus\source\repos\HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02\05_Generics>git push origin implementasi-generic-class
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.52 KiB | 779.00 KiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/rengganistan/HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02.git
c593015..65824ec implementasi-generic-class -> implementasi-generic-class

C:\Users\asus\source\repos\HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02\05_Generics>git checkout main
Switched to branch 'main'

C:\Users\asus\source\repos\HPL_Rengganis-Tantri-Pramudita_2311104065_515E-07-02\05_Generics>git branch
  implementasi-generic-class
  implementasi-generic-method
* main

```